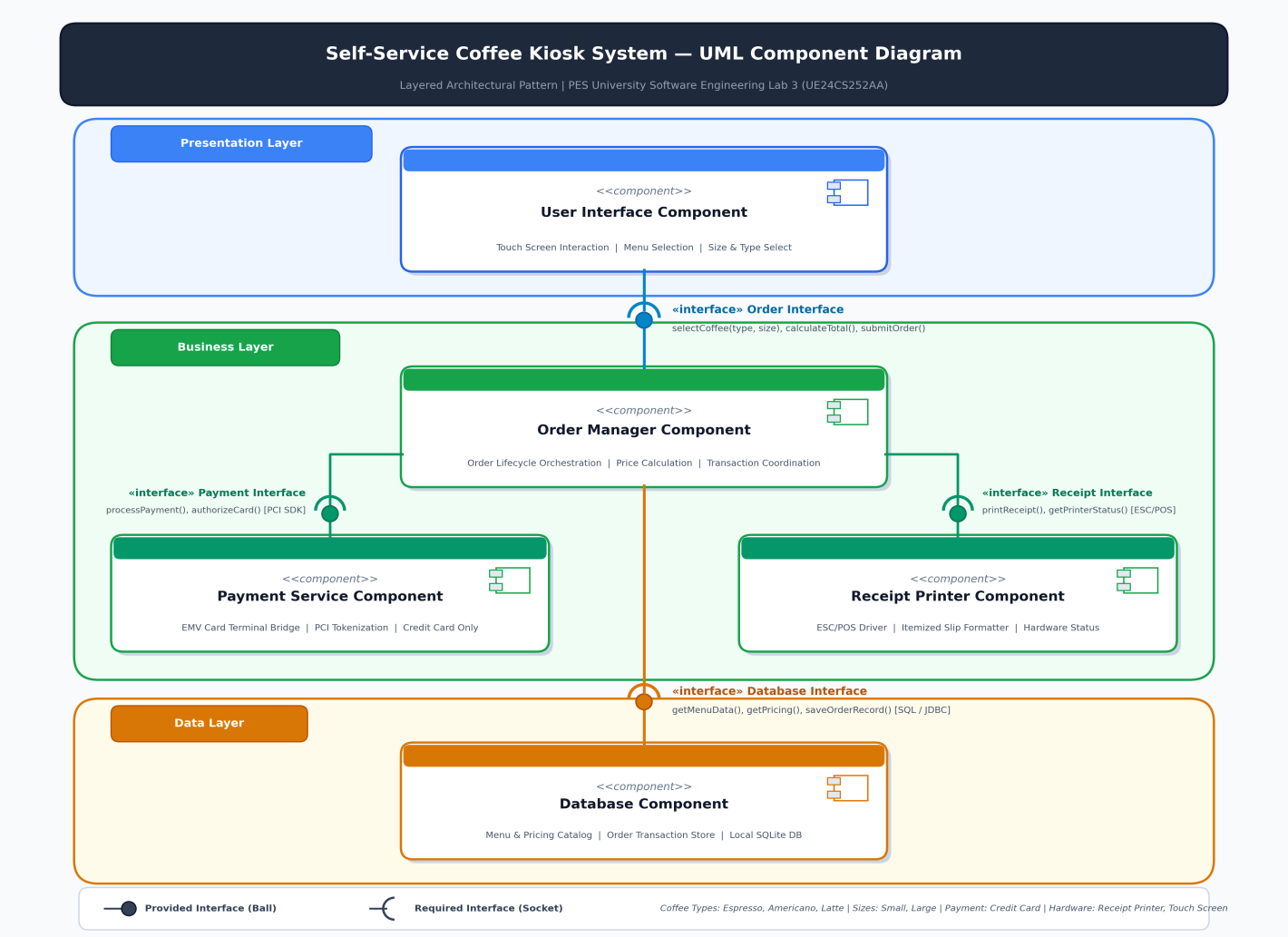
Architectural Specification & Design Document

Course: Software Engineering Lab (UE24CS252AA) — Lab 3: Component Modelling & Architectural Pattern Selection

Assigned System: Self-Service Coffee Kiosk System | **Student:** Sushruth S (PES1UG24CS486) | **Section:** 4-H



1. Architectural Choice & Justification

Architecture Selection: "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

- **Reason 1 — Hardware Modularity & Peripheral Isolation:** The kiosk directly interfaces with physical peripherals (capacitive touch screen, EMV card terminal, and ESC/POS thermal printer). Layered architecture isolates hardware-specific drivers into dedicated components, shielding core ordering domain logic from driver modifications.
- **Reason 2 — Elimination of Distributed Overhead:** The coffee kiosk is a single autonomous terminal serving customers sequentially. Adopting microservices would add networking latency and distributed failure points, whereas Layered architecture provides ultra-fast in-process IPC calls (<10ms).

2. Security Advantage & Performance Benefit

- **Security Advantage (PCI-DSS & Layer Isolation):** All credit card interactions are encapsulated strictly within Payment Service Component. The Touch Screen UI layer has zero visibility into raw cardholder data buffers or database storage, preventing memory-scraping attacks.
- **Performance Benefit:** Zero-network IPC overhead and pre-caching of static menu items (Espresso, Americano, Latte) and size pricing tables at startup guarantees instant touch responsiveness during peak rush hours.

3. System Components & Interfaces Summary

Component	Layer	Interfaces	Functionality
User Interface	Presentation	Requires: Order Interface	Touch screen interaction, menu & size selection
Order Manager	Business	Provides: Order Interface Requires: Payment, Receipt, DB	Order lifecycle orchestration, price computation

Payment Service	Business	Provides: Payment Interface	EMV card terminal bridge, PCI-DSS tokenization
Receipt Printer	Business (HW)	Provides: Receipt Interface	ESC/POS printer driver, itemized slip printing
Database	Data Layer	Provides: Database Interface	Menu catalog, size pricing, transaction audit log